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| AddSub     | This core is used to implement both adders and subtractors.                                                                    |
| AddSubnS   | N-stage pipelined adder or subtractor. Vivado HLS determines how many pipeline stages are required.                            |
| AddSub_DSP | This core ensures that the add or sub operation is implemented using a DSP48 (Using the adder or subtractor inside the DSP48). |
| DivnS      | N-stage pipelined divider.                                                                                                     |
| DSP48      | Multiplications with bit-widths that allow implementation in a single DSP48.                                                   |

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| Mul     | Combinational multiplier with bit-widths that exceed the size of a standard DSP48 macrocell.<br>Multipliers that can be implemented with a single DSP48 macrocell are mapped to the DSP48 core.                                                                                                                                                      |
| MulnS   | N-stage pipelined multiplier with bit-widths that exceed the size of a standard DSP48 macrocell.<br>Multiplications which are $\geq 10$ bits are implemented on a DSP48 macro cell. Multiplication lower than this limit are implemented using LUTs. Multipliers that can be implemented with a single DSP48 macrocell are mapped to the DSP48 core. |
| Mul_LUT | Multiplier implemented with LUTs.<br>Note: This only applies to C POD (plain old data) types. This cannot be used with Vivado HLS types (ap_int, ap_fixed, etc).                                                                                                                                                                                     |